Debojyoti Roy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.59
Birbhum Zilla School	Class XII (WBCHSE)	2023	91.80%
Suri P&CMM High School	Class X (WBBSE)	2021	97.10%

SKILLS

Languages: C, C++, Python, HTML, CSS, JavaScript, SQL

Frameworks and Libraries: ReactJS, ExpressJS, FastAPI, NumPy, Pandas, TensorFlow, Matplotlib, Seaborn, Scikit-Learn

Tools: Git, Github, MongoDB, Node.js, React.js, Postman, MySQL

Concepts and Technologies: Data Structures & Algorithms, OOPs

Soft Skills : Problem Solving, Communication, Critical Thinking, Time Management, Decision Making

PROJECTS

Text-to-Website Generator | Python, FastAPI, CrewAI, LLMs | Github

- Architected a **CrewAI multi-agent pipeline** in **FastAPI**, orchestrating specialized LLM agents to generate complete websites in **under 90 seconds**.
- Designed a **stateful feedback loop** for iterative code refinement, incorporating **safety guardrails** to block unsafe user inputs.
- Developed a **React.js frontend** with **Monaco Editor** and iframes for real-time code editing, rendering, and live UI previews.
- Implemented **one-click GitHub Pages deployment** using **OAuth 2.0** and GitHub APIs, with automated **README generation**.

AI Interview Preparation Platform | React, Node.js, Express.js, MongoDB, Multer, Zod, Google Gemini API | **Github**

- Built a **full-stack AI interview-prep tool** that generates a **personalized report** (match score, technical/behavioral questions, skill gaps, day-by-day prep plan) from a candidate's resume, self-description, and target job description
- Designed a **Node.js, Express.js + MongoDB** backend with **JWT authentication** (httpOnly cookies and a **token-blacklist** model for secure logout) and **RESTful APIs** for report generation, retrieval, and history
- Integrated **Google's Gemini API** with **Zod-validated structured JSON output** to parse resume PDFs and generate consistent, schema-conforming interview reports
- Built a **responsive React (Vite)** frontend with **protected routing**, **multipart file upload**, and a **multi-panel dashboard** for reviewing questions, skill gaps, and the prep roadmap

ShopNext -- Full-Stack MERN E-Commerce Platform | *React.js, Redux Toolkit, Context API, Node.js, Express.js, MongoDB, JWT, Nodemailer, Razorpay, Cloudinary* **| Github | Live Demo**

- Built a full-stack MERN e-commerce platform with **JWT authentication, OTP-based email verification, role-based access control, shopping cart, checkout, and order management.**
- Developed **30+ RESTful APIs** using Express.js and MongoDB for authentication, product CRUD, orders, payments, analytics, and user management.
- Integrated **Razorpay** for secure payment processing and **Cloudinary + Multer** for scalable product image uploads and management.
- Implemented **Redux Toolkit** for persistent cart state management and **React Context API** for authentication with protected routes and seamless user experience.

Customer Churn Prediction | Python, Pandas, Scikit-learn, XGBoost, SMOTE | **Github**

- Developed an end-to-end **customer churn classification pipeline** for 7K+ telecom customer records, including preprocessing, categorical encoding, and missing-value handling.
- Applied **SMOTE** to balance the churn classes and evaluated Decision Tree, Random Forest, and XGBoost using **5-fold cross-validation**.
- Achieved **84% cross-validation accuracy** with Random Forest and **77.9% test accuracy**, with a **0.58 F1-score** on the churn class.
- Implemented model persistence with **Pickle** and a reusable inference pipeline for generating churn predictions and probabilities on new customer profiles.

RELEVANT COURSES

CS Courses: Introduction To Programming, Data Structure and Algorithms

Maths Courses: Calculus I, Calculus II, Linear Algebra, Differential Equations, Intro To Probability, Transform Techniques, Complex Variables, Intro To Statistics

SCHOLASTIC ACHIEVEMENTS

- **400+** Data Structure and Algorithm Problems solved in Leetcode

EXTRACURRICULAR

- Member of IITH music club Vibes